

**FINANCIALIZATION EFFECT ON FIRM PERFORMANCE AND
ROLE OF FIRM SIZE IN FAMILY-OWNED PUBLIC LISTED
COMPANY IN INDONESIA**

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Abstract

Family business play an important role in Indonesia national economy as one of the main contributors of national GDP. However, there is a gap in understanding the financialization of family business in Indonesia, particularly on the impact of debt financing to the family business performance on profitability and market value. Therefore, this research study will contribute significantly to the practice of taking a financial leverage which will improve the performance of family business in Indonesia.

This study is an explanatory quantitative research study, and cross-sectional study for period 2016-2018. Using data from 412 listed company in IDX over the period, the study will elaborate the financial leverage effect on firm performance with contingent role on firm size. Further, this research uses 3 measure of financial leverage LTDTA, STDTA, and TDTA, and 3 measure of firm performance ROA, ROE, and market performance Tobin's Q.

Using statistical analysis, this study revealed that the non-family firm outperformed the family firm in term of accounting performance and market performance and suggested that family firm tend to utilize less debt compare to non-family firm. Subsequently, this study finds that there is a negative relationship between financial leverage and accounting performance of ROA, and a positive relationship between financial leverage and market performance of Tobin's Q.

Furthermore, this study implies that there is a contingent effect of firm size on the relationship between financial leverage and accounting performance of ROA. It is suggested that the negative effect of financial leverage on accounting performance of ROA is more prominent in small-sized company. It reduces the effect of the negative relationship on large-sized companies and show that financial leverage has a positive effect on market performance of Tobin's Q, more prominent in small-sized companies.

But because increased debt causes increased fragility, this research includes the calculation of industry benchmarks for maximum leverage for 8 different industries listed on the IDX. This maximum leverage can be used as a benchmark for each sector in using debt to generate better accounting performance of ROA.

1. INTRODUCTION

1.1 Background of Study

Family business has been always become an interesting topic for research, especially in Indonesia. In fact, Family Business has dominated the business landscape in Indonesia and spread into various industry and organizations (Tirdasari & Dhewanto, 2012). This research study will be conducted for family firms listed in Indonesia Stock Exchange (IDX). It has been shown that family business in Indonesia occupied of approximately 50% of listed company and 32% of total market capitalization in 2011. Moreover, Family business in Indonesia consider outperformed among other listed company in Indonesia after global economy crises 2008 (Wan & Redman, 2011).

In terms of business growth and contribution to the national economy, family businesses in Indonesia have shown a much stronger growth compare to global average growth in 2014. Further, family firms in Indonesia consist of approximately 51% of listed company and contribute at least 25% of Indonesia GDP (PwC, 2014). Moreover, the global business guide Indonesia consulting firm has reported that 95% of local business in Indonesia were family-owned business (Global Business Guide Indonesia, 2016). Eventually, most of the big companies in the world are come from a family firm, and found that one third of S&P 500 firms are owned by families (Anderson & Reeb, 2003).

Although there is already extensive literature on family businesses, this research will focus on the impact of financial leverage on the performance of family-owned company which is already listed in the IDX. Other studies on family businesses mainly focused on the relationship between the family ownership to the firm performance (Kim & Gao, 2013; Wistawan et al., 2015; Juanda & Jalaluddin, 2018), and how family control affects the company performance (Allouche et al., 2008; Kim & Gao, 2013; Yopie & Itan, 2016). Also, there are many extant literature on the effect of corporate governance on family business performance (Dyer, 1989; Achmad, 2007; Brenes et al., 2011).

However, there is still lack of study on the financial area of family business in Indonesia, particularly on the effect of debt financing to the family business performance on profitability and market value. Therefore, this research will contribute significantly to the practice of taking a financial leverage (debt) which may improve the profitability of family businesses in Indonesia.

There is two-way direction on the result of study on financial leverage impact on family business performance. First direction is to support the positive relationship between financial leverage on firm performance (Detthamrong et al., 2017; Goel et al., 2015; Rehman, 2013), and the other way is the proponent of negative relationship between financial leverage in firm performance (Salawu, 2007; González, 2013; Vithessonthi & Tongurai, 2015a, 2015b; Jermias & Yigit, 2019). Consequently, the relationship between financial leverage and firm performance in family-owned business are still unclear whether positive or negative relationship. Therefore, this study will confirm the result on the effect of financial leverage to firm performance in Indonesia family-owned public listed company.

The study of financial leverage mainly just to recognize the effect of the leverage to the firm performance. Therefore, there is a lack of study on industry-specific benchmarking on maximum leverage which will lead to positive company performance. Further, previous study on moderating effect of size of company on the relationship of financial leverage and firm performance only focusing on non-family businesses (Detthamrong et al., 2017; Ibhagui & Olokoyo, 2018; Ramli et al., 2019). Thus, the past relationship has ignored the contingent effect of size of the company related to financial leverage and firm performance in family-owned business.

1.2 Research Objective

This research study will be addressing following research objective:

- To determine the firm performance difference between family-owned company and non-family-owned company in Indonesia public listed company.
- To examine the relationship between financial leverage and firm performance in family-owned public listed company in Indonesia.
- To identify the contingent effect of size of the company to the relationship between financial leverage and firm performance
- To establish the industry-specific benchmark on maximum leverage which lead to better performance in family-owned public listed company in Indonesia.

The scope of study of this research study on family business will be described as following:

- This study will be focusing on family-owned business listed in Indonesia Stock Exchange (IDX).
- This study will be based on the last three years financial report (2016-2018) for the family-owned business listed in Indonesia Stock Exchange (IDX).

- This study will be based on small and large company which will be defined by the company's assets.
- This study will define a maximum leverage which will be determined by industry-specific within the family-owned public listed company in Indonesia

This paper will be organized as follows: section 1 will contain the introduction and motivation of the study. Section 2 will elaborate the literature review and the development of the hypotheses. Section 3 will describe the research methodology indicating the variables and techniques chosen to test the initial hypotheses. Section 4 deep dive into the analyses and findings. Finally, Section 5 will conclude the study with some final thoughts on the findings and their implications.

2. LITERATURE REVIEW

2.1 Family Business

Family business are firms which are classified such when the founding family members or descendent are holding shares or present in the board of directors position (Anderson & Reeb, 2003). There are many studies that examined the relationship between family firms and their performance. Some studies show that firm performance is improved when the founding family members is involved in the management (M. González et al., 2012; Lee, 2006). Others show that family involvement also leads to high performance (González-Cruz & Cruz-Ros, 2016). Also, family firms that are family-led tend to perform better than those are nonfamily-led (Dau et al., 2018).

From an organizational perspective, family businesses outperform non-family businesses alluding that family ownership is an effective organizational structure (Anderson & Reeb, 2003; Allouche et al., 2008). Moreover, family-owned businesses significantly have better performance than state-owned companies (Ding et al., 2008). A study showed that family firms have 7% higher valuation (Tobin's Q) and 16% higher profitability (ROA) relatively compared to other non-family firms (Maury, 2006). However, other conflicting studies show when family-owned companies have lower financial performance compare to non-family companies, family ownership has shown negative effects to the market value of the company (Bambang & Hermawan, 2012). This research will investigate at a deeper level if family businesses outperform non-family businesses, and if so what are the factors influential to their performance.

2.2 Financial Leverage

In terms of financial leverage, family businesses tend to have low debt to equity ratios (Gallo & Vilaseca, 1996). Further, when the family members join the management, they tend to be more risk averse, resulting in companies with lower debt levels. However, the needs for finance growth without losing control makes debt attractive but family businesses tend to favor internal sources of finance (Cleary & Alderighi, 2018).

Based on capital structure theory, many studies have been done on the relationship between financial leverage and company performance. Financial leverage can be defined as a relationship between borrowed fund and owner's fund in the capital structure of the company. It includes debt, common and preferred equity that use to finance the company assets, operational and financial growth (Goel et al., 2015). One of the researches mentioned that a moderate debt taking will lead to gain profit margin. However, after certain point, additional debts will lead to significant sales underperformance (Campello, 2006). It means that financial leverage is acceptable within business practice until certain limit of leverage.

However, most studies found that a firm's leverage has a negative impact on the firm's accounting performance. In contrary, the leverage measure has a positive and highly significant relationship with the firm's market performance (Olokoyo, 2013). Furthermore, the negative impact of leverage on firm performance is more prominent and significant within small-sized company and diminish as a firm grows in size (Ibhagui & Olokoyo, 2018). Then, larger firms have a higher financial leverage compare to smaller firms. It is suggested that the larger firms are more likely to use more debt financing than equity financing (Detthamrong et al., 2017).

Other studies show a negative result on firm's profitability associated with financial leverage (Salawu, 2007; González, 2013; Vithessonthi & Tongurai, 2015a, 2015b; Jermias & Yigit, 2019). Further, it was mentioned that company will exhaust their internal sources of funds before taking outsource funding, it is implied that there is a negative correlation between debt and profitability (Campello, 2006). However, there are also many studies describing the positive relationship on the impact of financial leverage to the firm performance (Rehman, 2013; Goel et al., 2015; Detthamrong et al., 2017).

2.3 Underpinning Theory (Theory of Capital Structure)

The underpinning theory of capital structure has been extensively discussed. Modigliani and Miller in 1958 has defined cost of capital as a fund that used to acquire assets which yields are uncertain. In which, the capital can be obtained from different media such as: debt instruments

and equity issuance (Modigliani & Miller, 1958). Further, other studies state that incentive effects of the debt are large relative to the incentive of equity. Firms like conglomerates should be characterized by utilizing lower debt. Conversely, in the industry where there is a freedom of market, the company should have more debt financing (Jensen & Meckling, 1976). Subsequently, many researchers have conducted numerous studies regarding capital structure topics which study relationship between financial leverage to performance of the firms which has been explained in previous sub chapter of financial leverage.

2.4 Developing Hypotheses

As previously observed, the family firms that are family-led tend to perform better than those are nonfamily-led (Allouche et al., 2008; Anderson & Reeb, 2003; Dau et al., 2018). Moreover, Family business in Indonesia have outperformed other publicly listed companies in Indonesia for the period since the global economy crises of 2008 (Wan & Redman, 2011).

Also, family businesses tend to have low debt ratios (Gallo & Vilaseca, 1996). They tend to be more risk averse which is push the company towards lower debt levels (González et al., 2013). Further, there is a fact that family business tend to favor internal or traditional resource of finance (Cleary & Alderighi, 2018).

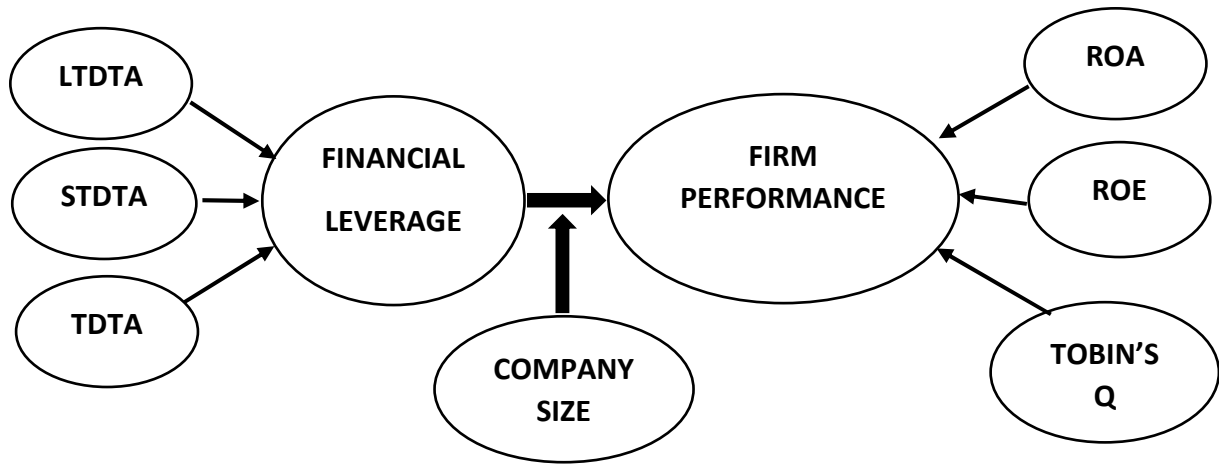
The literature review on firm's leverage suggest a negative impact on the firm's accounting performance. However, the leverage measure has a positive and highly significant relationship with the firm's market performance (Olokoyo, 2013). Further, it was mentioned that company will exhaust their internal sources of funds before taking outsource funding, it is implying that there is a negative correlation between debt and profitability (Campello, 2006).

Other studies provide a negative relationship on firm's profitability associated with financial leverage (Salawu, 2007; González, 2013; Vithessonthi & Tongurai, 2015a, 2015b; Jermias & Yigit, 2019). Other studies also mention that debt financing could increase the company value, because debt financing could reduce free cash flow of the enterprise (Yang et al., 2016). And that the effects of leverage is positive on the market value of the company (Tobin's Q) (Ibhagui & Olokoyo, 2018).

The negative impacts of leverage on firm performance is more prominent and significant within small-sized company and diminished as a firm grows (Ibhagui & Olokoyo, 2018), since the larger the firms more likely to generate more returns that give better performance (Ramli et al., 2019). Further, the measures of financial leverage are positively linked with market performance of Tobin's Q and the positive effect is much stronger in the small-sized company.

2.5 Conceptual Framework

Figure 1 Conceptual Framework



2.5.1 Dependent Variable

In this study, the firm's performance measure commonly used return on assets (ROA), return on equity (ROE), and Tobin's Q (Maury, 2006; Bunkanwanicha et al., 2008; Olokoyo, 2013; Yang et al., 2016). Profitability ratios that most suit to describe company's performance and commonly used by investor to measure profit is Return on Assets (ROA) (Surifah, 2013), other profitability ratios that commonly used is Return on Equity (ROE) (Allouche et al., 2008; Singapurwoko, 2013). Further, the market value of the company is measured by Tobin's Q, which indicates the ratio between market value and the replacement value of the company assets (Bunkanwanicha et al., 2008; Sumarsono, 2014).

ROA can be calculated by dividing Earnings Before Interest and Taxes (EBIT) by total assets (EBIT/Total Assets), then calculating ROE by dividing Earnings Before Interest and Taxes (EBIT) by total equity (EBIT/Total Equity), and calculating Tobin's Q by dividing market recapitalization by company total assets (Market Value/Total Assets).

2.5.2 Independent Variable

In this study, the financial leverage measure commonly used total debt ratio (TDTA), long term debt ratio (LTDTA), and short term debt ratio (STDTA) (Olokoyo, 2013; Yang et al., 2016). The debt rate can be measured by Total Debt over Total Assets (TDTA) (Allouche et al., 2008; Bunkanwanicha et al., 2008). Then, long term leverage can be measured by dividing long term debt to total assets (LTDTA) which can help to reduce the potential of reverse causality of performance and capital structure

(Campello, 2006). Eventually, The short term debt can be measured for defining financial leverage by dividing Short Term Debt by Total Assets (STDTA) which have a positive relationship with the company growth (Salawu, 2007)

2.5.3 Moderating Variable

The proposed moderating variable will be company size. Small-sized company will be defined when the company assets fall below the asset's median of samples, and large-sized company will be defined when the company assets exceed the asset's median of the samples (Hao & Song, 2016) .

There are many extant studies on family business. However, there is a lack of study on the financial leverage effect on family business performance specially in Indonesia. Based on theory of capital structure, this research study will determine financial leverage impact on the firm performance of family-owned business which is already listed in IDX (Indonesia Stock Exchange).

3. RESEARCH METHODOLOGY

3.1 Research Design

This research will be studying the relationship between financial leverage and firm performance in the family business environment in Indonesia. It will be conducting **cross-sectional studies** to understand the phenomenon of the financial leverage effect on firm performance in family-owned public listed company in Indonesia from year 2016 to 2018. This study took the last 3 years data in order to have a recent pictures of financial leverage practices in Indonesia family business.

3.2 Population and Sample

The sample of the research is based on 632 companies listed on the Indonesia Stock Exchange (IDX) from 2016-2018 period (Indonesia Stock Exchange, 2019). Samples will be selected based on purposive sampling method which relies on family-owned companies listed on the IDX. Data will be gathered from secondary data of last three years (2016-2018) financial statement of family-owned company listed in IDX to ensure this research study has a recent pictures of financial leverage practices.

The selection of family-owned company will be based on at least 5% of shares owned by family members or the family members become Board of Directors or Board of Commissioners

(Allouche, Amann, Jaussaud, & Kurashina, 2008). This study only includes non-financial family-owned firms on the exchange.

3.3 Instruments and Measurements

3.3.1 Firm's Performance (Dependent Variables)

The firm's performance measures are the commonly used factors which are the Return on Assets (ROA), Return on Equity (ROE), and Tobin's Q (Bunkanwanicha et al., 2008; Maury, 2006; Olokoyo, 2013; Yang et al., 2016). Profitability ratios that is most suited to describe a company's performance and commonly used by investor to measure profit is its Return on Assets (ROA) (Surifah, 2013), other profitability ratios that are commonly used is Return on Equity (ROE) (Allouche et al., 2008; Singapurwoko, 2013). For the market value of the company, this study utilizes Tobin's Q, which indicates the ratio between market value and the replacement value of the company assets (Bunkanwanicha et al., 2008; Sumarsono, 2014).

ROA can be calculated by dividing Earnings Before Interest and Taxes (EBIT) by total assets (EBIT/Total Assets), then calculating ROE by dividing Earnings Before Interest and Taxes (EBIT) by total equity (EBIT/Total Equity), and calculating Tobin's Q by dividing market recapitalization by company total assets (Market Value/Total Assets).

3.3.2 Financial Leverage (Independent Variable)

In this research, the financial leverage measures are the commonly used factors, namely total debt ratio (TDTA), long term debt ratio (LTDTA), and short term debt ratio (STDTA) (Olokoyo, 2013; Yang et al., 2016). The debt rate is measured by Total Debt over Total Assets (TDTA) (Allouche et al., 2008; Bunkanwanicha et al., 2008), and the long term leverage is measured by dividing long term debt to total assets (LTDTA) which is able to reduce the potential of reverse causality of performance and capital structure (Campello, 2006). Finally, the short term debt is measured by dividing Short Term Debt by Total Assets (STDTA) which have a positive relationship with the company growth (Salawu, 2007).

3.3.3 Company Size (Moderating Variable)

The proposed moderating variable is the size of the company. Small-sized companies are defined as company assets being below the asset's median of samples, and large-

sized companies are defined as company assets exceeding the asset's median of the samples (Vithessonthi & Tongurai, 2015b; Hao & Song, 2016).

3.4 Data Collection Procedures

As for data collection, the sample of the study will be divided into 2 groups: family firms and non-family firms. Data will be collected from audited financial report of public listed company listed in IDX (Indonesia Stock Exchange). The main advantages of using audited financial report is that the data is reliable, accurate, consistent across periods. Family firms' criteria is minimum of 5% ownership belong to family members or the family members become board directors or commissioners.

Samples in this study will include all industry except financial firms. Financial report will be access directly from the IDX website and company website. Incomplete financial report will be removed and to be excluded from the research study. The financial report of year 2016 to 2018 is used for this research, this study wants to have a recent pictures of financial leverage practices in family-owned business listed in IDX.

There are 543 non-financial companies listed in IDX, 131 companies were excluded from the study due to incomplete financial reports. Hence, only 412 companies qualified for this study, 189 companies are family firms and 223 companies are non-family firms. As suggested by previous study, this study identified family members by their surname and checked for family affiliation in annual report (Damayanti et al., 2018).

3.5 Data Analysis

3.5.1 Descriptive Statistics

Descriptive statistics analysis will be conducted to describe the statistic figures of the variables within the research study such as mean, medians and standard deviation (Jermias & Yigit, 2019). This research used IBM SPSS 25 to run descriptive statistics analysis.

3.5.2 Correlation Analysis

Pearson correlation analysis will be used to test the correlation and multicollinearity tests between independent and dependent variables. This correlation analysis will obtain a preliminary view of the bivariate association between variables. Further, it will help to objectively determine a similarity between variables and ensure this study used appropriate variables for regression linear analysis (Ibhagui & Olokoyo, 2018).

3.5.3 Regression Linear Analysis

Regression linear analysis will be conducted to measure the relationship between financial leverage and firm performance in Indonesia family business. To measure the contingent effect of company size to the financial leverage relationship to firm performance, the samples will be divided into 2 (two) subsamples and do the regression analysis towards both sub-samples to identify the contingent effect of company size (Bambang & Hermawan, 2012).

Accordingly, a functional relationship between firm performance (PER) and the explanatory variables of financial leverage (LEV) is shown below:

$$PER = f(LEV)$$

With:

$$(PER)' = (ROA, ROE, Tobin's Q)'$$

$$(LEV)' = (TDTA, LTDTA, STDTA)'$$

PER represents the different measures of firm performances (*ROA, ROE, Tobin's Q*), and *LEV* represent the different measures of financial leverage (*TDTA, LTDTA, STDTA*). Further, the relationship between the component of *PER* and the different explanatory variables can be rewrite implicitly as following:

$$ROA = f(TDTA, LTDTA, STDTA, v)$$

$$ROE = f(TDTA, LTDTA, STDTA, \mu)$$

$$TOB Q = f(TDTA, LTDTA, STDTA, v)$$

With: v, μ, v are terms of error. Hence, the analytical data model to be tested in this study consist of equations which structured as following:

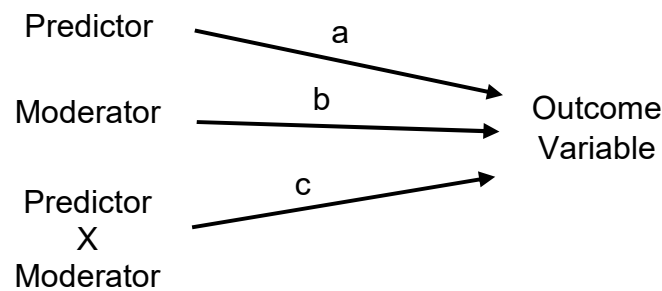
$$y = \alpha + \beta x + \mu$$

Where: y is a vector of dependent variables (PER), such that $(y)' = (ROA, ROE, Tobin's Q)'$. and X is a vector of independent variables (LEV), such that $(X)' = (TDTA, STDTA, LTDTA)'$.

3.5.4 Moderator Analysis

In general term, moderator is a variable that affect the direction and/or strength of relation between independent and dependent variable. The essential properties of a moderator variable are summarized in below figure:

Figure 2 Moderator Model



The moderator effect is supported when the path c interaction is significant. There may also a significant effect of the main path of the predictor and moderator (path a and b). However, these are not directly relevant concept to testing for moderator effect (Baron & Kenny, 1986).

To test for moderator effect, regression linear analysis can be used for testing moderating variables (predictor x moderator) relationship with the outcome variable. This study will use IBM SPSS 25 for testing regression linear analysis for moderating variables.

3.5.5 Maximum Leverage

Maximum leverage can be defined as a maximum value of leverage that could lead to a positive result of the company performance. Based on the regression linear analysis result, the linear function of the relationship between financial leverage and company performance as already explained in sub-chapter 3.6.3 is describe as following:

$$y = \alpha + \beta x + \mu$$

Assume that term of error is 0, then the linear function become:

$$y = \alpha + \beta x$$

Then, the maximum leverage can be achieved when the function of company performance is exceeding 0 ($y > 0$), then the linear function become:

$$y > 0, \alpha + \beta x > 0$$

$$x < \left| \frac{\alpha}{\beta} \right|$$

Therefore, the maximum leverage can be achieved by dividing an absolute number of constant coefficient alpha and unstandardized coefficient of beta.

4. RESULTS AND DISCUSSION

4.1 Descriptive Statistics

Based on the firm type criteria of family firm and non-family firm, this research has selected 223 non-family firms and 189 family firms from 8 different industries listed in Indonesia Stock Exchange (IDX). In Table 1, it shows a descriptive statistic for 2 groups of family firms and non-family firms for company leverage and performance ratio.

Table 1 Descriptive Statistics (All Variables)

	Firm Type	N	Minimum	Maximum	Mean	Std. Deviation
LTDTA	Family Firms	189	0.0030	7.0487	0.2660	0.5279
	Non Family Firms	223	0.0022	11.1491	0.3139	0.9353
STDTA	Family Firms	189	0.0023	10.2306	0.3165	0.7628
	Non Family Firms	223	0.0018	5.1904	0.3583	0.5847
TDTA	Family Firms	189	0.0404	17.2792	0.5824	1.2485
	Non Family Firms	223	0.0125	16.3395	0.6722	1.4005
ROA	Family Firms	189	-9.8083	0.3338	-0.0227	0.7284
	Non Family Firms	223	-0.7486	1.0857	0.0380	0.1565
ROE	Family Firms	189	-101.0285	1.9283	-0.5238	7.4019
	Non Family Firms	223	-262.6533	61.5718	-0.6807	18.1996
TOBIN Q	Family Firms	189	0.2927	28.3920	1.7031	2.5600
	Non Family Firms	223	0.2273	138.2385	2.8599	11.4469
Valid N (listwise)	Family Firms	189				
	Non Family Firms	223				

From the mean comparison, it shows that the mean of the non-family firm ROA (0.038) is greater than the mean of family firm ROA (-0.0227). On the other hand, the mean of non-family firm ROE (-0.6807) is indicated to be lower than the mean of family firm ROE (-0.5238). While the mean of non-family firm Tobin's Q (2.8599) is greater than the mean of family firm Tobin's Q (1.7031).

In general, there is an indication of which non-family firm have a better performance compare to family firm. Therefore, it can be concluded that the performance of non-family firm is outperformed compare to family firm.

This summary of statistics has revealed several important findings for both independent and dependent variables. Profitability performance of ROA for both family and non-family firm shows relatively low performance with average value of -2.27% for family firms and 3,8% for non-family firms. It is reflecting low efficiency for both group of companies in creating profits relative to their total assets.

Moreover, profitability performance of ROE for both family and non-family firms shows low and negative performance with average value of -52.38% for family firm and -68.07% for non-family firm. Negative performance of ROE can be obtained from either negative net profit or negative equity. Negative net profit shows low efficiency for both groups of company in generating profits. While negative equity shows relatively high numbers of liabilities to fund their assets compare to their equity.

Furthermore, market performance of Tobin's Q for both family and non-family firms shows high performance with average value of 1.7 times for family firms and 2.8 times for non-family firms. It indicates that both group of companies has been valued above their book value (over-valued) which can be resulted from share price increase and high rate of debt. On the other words, their market performance has been exceeding their internal performance.

In terms of debts, both of group companies of family and non-family considered have a low and medium rate of debts. Average value of 26,6% for family firms LTDTA and 31.39% for non-family firms STDA show low rate of long-term debts to fund their assets. Other than that, average value of 31.65% for family firms STDTA and 35.83% for non-family firms STDTA also show low rate of short-term debts to fund their assets. While average value of 58.24% for family firms TDTA and 67.22% for non-family firms TDTA show medium rate of total debts to fund their assets.

In general, family firms tend to have less debt rate compare to non-family firms. It may indicate that family firms tend to be more avoiding debts compare to non-family firms.

4.2 Correlation Matrix

Correlation matrix between variables is presented in table 2. Pearson correlation analysis has been used to identify any correlation which exist among variables. This correlation matrix shows a significant negative relationship between ROA and financial leverage variables which consist of three measurements of LTDTA, STDTA, and TDTA. The correlation range is between -17.9% to -14.4% with significant result at 99% confidential level. This result indicates that financial leverage has a negative impact on accounting performance of ROA.

Table 2 Pearson Correlations Family Firms Matrix

	LTDTA	STDTA	TDTA	ROA	ROE	TOBIN Q
LTDTA	1.000	0.867**	0.952**	-0.144*	-0.004	0.437**
STDTA	0.867**	1.000	0.978**	-0.179*	-0.030	0.460**
TDTA	0.952**	0.978**	1.000	-0.170*	-0.020	0.466**
ROA	-0.144*	-0.179*	-0.170*	1.000	0.977**	-0.055
ROE	-0.004	-0.030	-0.020	0.977**	1.000	0.013
TOBIN Q	0.437**	0.460**	0.466**	-0.055	0.013	1.000

** . Correlation is significant at the 0.01 level (2-tailed).

* . Correlation is significant at the 0.05 level (2-tailed).

On the other hand, the correlation between ROE and financial leverage variables is not significant with negative relationship between -3% to -0.4% correlation value. It may imply that there is no significant relationship between financial leverage and accounting performance of ROE.

Furthermore, the financial leverage variables are positively correlated with Tobin's Q with correlation range between 43.7% to 46.6% and give significant result at 95% confidential level. This result indicates that financial leverage has a positive influence on market performance Tobin's Q.

4.3 Linear Regression

Tables 3, 4, and 5 are presenting the regression analysis result between financial leverage variables and company performance of ROA, ROE and Tobin's Q. In line with correlation matrix in previous sub-chapter, there is a significant negative relationship between financial leverage variables and accounting performance of ROA with p-value below 0.05 ($p < 0.05$) at 95% confidential level and negative numbers of standardized coefficient beta (Table 3). It means that higher level of leverage will lead to low level of ROA.

Table 3 Regression Analysis Family Firms (Dependent Variable ROA)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
		B	Std. Error								
LTDTA	(Constant)	0.030	0.059		0.514	0.608	0.144	0.021	0.016	0.7227190	3.979
	LTDTA	-0.199	0.100	-0.144	-1.995	0.048					
STDTA	(Constant)	0.031	0.057		0.554	0.580	0.179	0.032	0.027	0.7185635	6.194
	STDTA	-0.171	0.069	-0.179	-2.489	0.014					
TDTA	(Constant)	0.035	0.058		0.609	0.543	0.170	0.029	0.024	0.7196816	5.594
	STDTA	-0.099	0.042	-0.170	-2.365	0.019					

However, the regression result between financial leverage variables and accounting performance of ROE shows a non-significant negative relation as p-value exceed 0.05 ($p > 0.05$) at 95% confidential level and negative numbers of standardized coefficient beta (table 4). It means that there is no significant relationship between financial leverage measurement and accounting performance of ROE.

Table 4 Regression Analysis Family Firms (Dependent Variable ROE)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
		B	Std. Error								
LTDTA	(Constant)	-0.510	0.605		-0.844	0.400	0.004	0.000	-0.005	7.4216103	0.002
	LTDTA	-0.051	1.025	-0.004	-0.049	0.961					
STDTA	(Constant)	-0.431	0.584		-0.738	0.461	0.030	0.001	-0.004	7.4183036	0.169
	STDTA	-0.292	0.709	-0.030	-0.411	0.681					
TDTA	(Constant)	-0.455	0.596		-0.764	0.446	0.020	0.000	-0.005	7.4201899	0.074
	STDTA	-0.118	0.433	-0.020	-0.272	0.786					

In table 5, the regression result shows a significant positive relationship between financial leverage variables and market performance of Tobin's Q at 99% confidential level with all p value below 0.01 ($p < 0.01$) and positive numbers of standardized coefficient beta. It means that higher leverage will lead to higher market performance of Tobin's Q.

Table 5 Regression Analysis Family Firms (Dependent Variable Tobin's Q)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
		B	Std. Error								
LTDTA	(Constant)	1.139	0.188		6.055	0.000	0.437	0.191	0.187	2.3085119	44.187
	LTDTA	2.120	0.319	0.437	6.647	0.000					
STDTA	(Constant)	1.215	0.180		6.764	0.000	0.46	0.212	0.207	2.2791653	50.179
	STDTA	1.544	0.218	0.460	7.084	0.000					
TDTA	(Constant)	1.147	0.182		6.287	0.000	0.466	0.217	0.213	2.2712244	51.84
	STDTA	0.955	0.133	0.466	7.200	0.000					

4.4 Moderating Effect

4.4.1 Moderator Analysis

Tables 6, 7, and 8 present moderator analyses using regression linear analysis. The tables present the influence of the moderating variable of firm size on financial leverage and firm performance relationships. Table 6 shows that moderating variables of firm size (Moderator LTDTA x SIZE) give a significant impact to LTDTA and ROA relationship as the p-value is below 0.1 ($p = 0.062$) which is significant at 90% confidential level.

Table 6 Moderator Analysis Family Firm (Dependent Variable ROA, Moderator Variable Firm Size)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F	Model Sig.
		B	Std. Error	Beta								
LTDTA	(Constant)	0.068	0.164		0.413	0.680	0.144	0.021	0.010	0.7247145	1.964	0.143
	Moderator LTDTA x SIZE	-0.175	0.093	-0.141	-1.875	0.062						
	Firm Size	-0.014	0.109	-0.010	-0.132	0.895						
STDTA	(Constant)	0.133	0.163		0.819	0.414	0.217	0.047	0.037	0.7148391	4.606	0.011
	Moderator STDTA x SIZE	-0.191	0.064	-0.213	-2.965	0.003						
	Firm Size	-0.051	0.104	-0.035	-0.485	0.628						
TDTA	(Constant)	0.104	0.162		0.638	0.524	0.200	0.040	0.030	0.7175550	3.869	0.023
	Moderator TDTA x SIZE	-0.111	0.041	-0.197	-2.705	0.007						
	Firm Size	-0.024	0.106	-0.017	-0.227	0.820						

Moreover, the moderator variables of firm size (Moderator STDTA x SIZE) give a significant impact on STDTA and ROA relationship as the p-value is below 0.01 ($p = 0.003$) which is significant at 99% confidential level. Further, the moderator variable of firm size (Moderator TDTA x SIZE) give a significant impact on TDTA and ROA relationship as the p-value is below 0.01 ($p = 0.007$) which is significant at 99% confidential level.

Table 7 Moderator Analysis Family Firm (Dependent Variable ROE, Moderator Variable Firm Size)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F	Model Sig.
		B	Std. Error	Beta								
LTDTA	(Constant)	0.898	1.677		0.535	0.593	0.066	0.004	-0.006	7.4255676	0.402	0.670
	Moderator LTDTA x SIZE	-0.011	0.954	-0.001	-0.012	0.991						
	Firm Size	-0.967	1.121	-0.065	-0.862	0.390						
STDTA	(Constant)	1.080	1.684		0.641	0.522	0.094	0.009	-0.002	7.4087118	0.827	0.439
	Moderator STDTA x SIZE	-0.615	0.668	-0.067	-0.921	0.358						
	Firm Size	-0.915	1.082	-0.062	-0.846	0.399						
TDTA	(Constant)	0.959	1.678		0.571	0.568	0.078	0.006	-0.005	7.4187308	0.574	0.564
	Moderator TDTA x SIZE	-0.247	0.422	-0.043	-0.586	0.559						
	Firm Size	-0.873	1.095	-0.059	-0.797	0.426						

Table 7 shows that moderating variable of firm size give a non-significant influence for all financial leverage variables and ROE relationship as the p-value is exceed 0.05 ($p > 0.05$) at 95% confidential level. However, table 8 shows that moderating variables of firm size give a significant influence on financial leverage variables and Tobin's Q relationship as the p-value is below 0.01 ($p < 0.01$) at 99% confidential level.

Table 8 Moderator Analysis Family Firm (Dependent Variable Tobin's Q, Moderator Variable Firm Size)

Model	Predictors	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F	Model Sig.
		B	Std. Error	Beta								
LTDTA	(Constant)	2.263	0.533		4.250	0.000	0.401	0.161	0.152	2.3579321	17.799	0.000
	Moderator LTDTA x SIZE	1.785	0.303	0.410	5.894	0.000						
	Firm Size	-0.865	0.356	-0.169	-2.429	0.016						
STDTA	(Constant)	1.773	0.528		3.357	0.001	0.431	0.185	0.177	2.3229597	21.160	0.000
	Moderator STDTA x SIZE	1.348	0.209	0.427	6.437	0.000						
	Firm Size	-0.440	0.339	-0.086	-1.298	0.196						
TDTA	(Constant)	1.955	0.520		3.758	0.000	0.449	0.202	0.193	2.2997595	23.475	0.000
	Moderator TDTA x SIZE	0.889	0.131	0.450	6.785	0.000						
	Firm Size	-0.671	0.339	-0.131	-1.977	0.050						

4.4.2 Group Linear Regression

Tables 9, 10, and 11 present group regression analyses of family-owned company by firm size. It shows a comparison between small size and large size company in relation to financial leverage and firm performance relationship.

Table 9 Group Regression Analysis Family Firm by Firm Size (Dependent Variable ROA)

Model	Firm Size	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
			B	Std. Error								
LTDTA	Small Firm	(Constant)	0.061	0.010		6.349	0.000	0.848	0.719	0.716	0.0902701	253.459
		LTDTA	-0.204	0.013	-0.848	-15.920	0.000					
	Large Firm	(Constant)	-0.036	0.211		-0.168	0.867	0.014	0.000	-0.011	1.0600851	0.017
		LTDTA	-0.084	0.639	-0.014	-0.131	0.896					
STDTA	Small Firm	(Constant)	0.064	0.009		6.951	0.000	0.861	0.741	0.739	0.0866162	283.823
		STDTA	-0.143	0.009	-0.861	-16.847	0.000					
	Large Firm	(Constant)	0.152	0.157		0.964	0.338	0.199	0.040	0.029	1.0389264	3.556
		STDTA	-0.886	0.470	-0.199	-1.886	0.063					
TDTA	Small Firm	(Constant)	0.064	0.009		7.200	0.000	0.870	0.756	0.754	0.0841022	307.052
		TDTA	-0.087	0.005	-0.870	-17.523	0.000					
	Large Firm	(Constant)	0.381	0.257		1.483	0.142	0.200	0.040	0.029	1.0386791	3.599
		TDTA	-0.851	0.449	-0.200	-1.897	0.061					

In line with the moderator analyses results, the moderating variable of firm size show a significant influence on financial leverages and ROA relationship. Table 9 shows that small-sized company has a significant negative relation between financial leverage and ROA with p-value below 0.01 ($p < 0.01$) for all financial leverage variables at 99% confidential level. Then, the standardized coefficient beta for all financial leverage variables shows a negative result.

On the other hand, table 9 shows that large-sized company will reduce the influence on financial leverage and ROA as the standardized coefficient beta become lesser in numbers which shows decrease of negative impact of financial leverage and ROA.

Table 10 Group Regression Analysis Family Firm by Firm Size (Dependent Variable ROE)

Model	Firm Size	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
			B	Std. Error								
LTDTA	Small Firm	(Constant)	-0.060	0.124		-0.485	0.629	0.028	0.001	-0.009	1.1751791	0.075
		LTDTA	-0.046	0.167	-0.028	-0.274	0.785					
	Large Firm	(Constant)	-1.124	2.163		-0.519	0.605	0.005	0.000	-0.012	10.8471470	0.002
		LTDTA	0.291	6.534	0.005	0.044	0.965					
STDTA	Small Firm	(Constant)	-0.066	0.125		-0.526	0.600	0.014	0.000	-0.010	1.1755107	0.019
		STDTA	-0.016	0.116	-0.014	-0.139	0.890					
	Large Firm	(Constant)	0.755	1.619		0.467	0.642	0.166	0.028	0.016	10.6961299	2.448
		STDTA	-7.569	4.838	-0.166	-1.564	0.121					
TDTA	Small Firm	(Constant)	-0.063	0.125		-0.505	0.614	0.020	0.000	-0.010	1.1753940	0.039
		TDTA	-0.014	0.069	-0.020	-0.197	0.844					
	Large Firm	(Constant)	2.451	2.653		0.924	0.358	0.155	0.024	0.013	10.7154633	2.129
		TDTA	-6.753	4.629	-0.155	-1.459	0.148					

Table 10 shows a non-significant influence on moderating variable of firm size in relation to financial leverage and ROE relationship as the p-value exceed 0.05 ($p > 0.05$) at 95% confidential level for both small-size and large-sized company and for all

financial leverages variables relation to ROE. Hence, there is no moderating effect of firm size on the relation between financial leverage and ROE.

Table 11 Group Regression Analysis Family Firm by Firm Size (Dependent Variable TOBIN'S Q)

Model	Firm Size	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	R	R Square	Adjusted R Square	Std. Error of the Estimate	F
			B	Std. Error								
LTDTA	Small Firm	(Constant)	1.264	0.316		4.006	0.000	0.481	0.232	0.224	2.9822952	29.829
		LTDTA	2.313	0.423	0.481	5.462	0.000					
	Large Firm	(Constant)	1.863	0.189		9.835	0.000	0.217	0.047	0.036	0.9498324	4.264
		LTDTA	-1.182	0.572	-0.217	-2.065	0.042					
STDTA	Small Firm	(Constant)	1.228	0.316		3.881	0.000	0.486	0.236	0.229	2.9727499	30.658
		STDTA	1.618	0.292	0.486	5.537	0.000					
	Large Firm	(Constant)	1.571	0.147		10.676	0.000	0.040	0.002	-0.010	0.9723203	0.137
		STDTA	-0.163	0.440	-0.040	-0.370	0.712					
TDTA	Small Firm	(Constant)	1.223	0.315		3.881	0.000	0.492	0.242	0.234	2.9617197	31.625
		TDTA	0.983	0.175	0.492	5.624	0.000					
	Large Firm	(Constant)	1.923	0.236		8.138	0.000	0.194	0.038	0.026	0.9546255	3.360
		TDTA	-0.756	0.412	-0.194	-1.833	0.070					

Table 11 shows that small-sized company has a significant positive relation between financial leverage and Tobin's Q with p-value below 0.01 ($p < 0.01$) for all financial leverage variables at 99% confidential level. The standardized coefficient beta for all financial leverage variables is positive.

While table 11 shows that large-sized company will reduce the influence on financial leverage and Tobin's Q as the standardized coefficient beta become negative which indicates a decrease of positive impact of financial leverage and Tobin's Q.

4.5 Industry Benchmark for Maximum Leverage

4.5.1 Industry Sectors Descriptive Statistics

This investigation has selected 189 family firms from 8 different sectors listed in IDX, which are: Agriculture, Mining, Basic Industry and Chemicals, Miscellaneous Industry, Consumer Goods Industry, Property, Real Estate, and Building Construction, Infrastructure, Utilities, and Transportation, Trade, Services, and Investment.

Table 12 shows a descriptive statistic for 8 different sectors. Average value of LTDTA is range between 13.8% and 56.49% for Consumer Goods Industry and Infrastructure, Utilities, and Transportation, respectively. Subsequently, Average value of STDTA is range between 4.6% and 71.3% for Property, Real Estate, and Building Construction, and Infrastructure, Utilities, and Transportation, respectively.

Table 12 Descriptive Statistics by Industry Sectors

Sectors	Variables	N	Minimum	Maximum	Mean	Std. Deviation
Agriculture	LTDTA	5	0.1899	0.5165	0.3254	0.1342
	STDTA	5	0.0902	0.8244	0.3094	0.3022
	TDTA	5	0.3288	1.0143	0.6347	0.2449
	ROA	5	-0.1921	0.0675	-0.0228	0.1114
	ROE	5	-0.5631	1.3356	0.2231	0.6887
	TOBIN Q	5	0.7153	1.2087	0.9882	0.1783
	Valid N (listwise)	5				
Mining	LTDTA	17	0.0126	0.5885	0.2920	0.1708
	STDTA	17	0.0565	0.6039	0.2280	0.1367
	TDTA	17	0.1552	0.8703	0.5200	0.2069
	ROA	17	-9.8083	0.3338	-0.5137	2.3968
	ROE	17	-101.0285	0.5630	-5.8174	24.5359
	TOBIN Q	17	0.4734	3.4257	1.1574	0.6768
	Valid N (listwise)	17				
Basic Industry and Chemicals	LTDTA	29	0.0128	0.8666	0.1832	0.1838
	STDTA	29	0.0777	0.8220	0.3395	0.2087
	TDTA	29	0.1205	1.1106	0.5227	0.2684
	ROA	29	-0.1088	0.1359	0.0339	0.0582
	ROE	29	-3.7416	0.4282	-0.1121	0.7890
	TOBIN Q	29	0.3810	2.8829	1.1120	0.6080
	Valid N (listwise)	29				
Miscellaneous Industry	LTDTA	18	0.0107	0.6630	0.1946	0.2115
	STDTA	18	0.0861	1.0083	0.4153	0.2494
	TDTA	18	0.1008	1.6664	0.6099	0.3482
	ROA	18	-0.1340	0.2265	0.0315	0.0854
	ROE	18	-10.8274	0.3975	-0.4832	2.5853
	TOBIN Q	18	0.6990	4.7028	1.3582	0.9355
	Valid N (listwise)	18				
Consumer Goods Industry	LTDTA	17	0.0117	0.4401	0.1380	0.1381
	STDTA	17	0.0708	0.4880	0.2445	0.1192
	TDTA	17	0.0826	0.6196	0.3826	0.1781
	ROA	17	-0.0586	0.2060	0.0768	0.0829
	ROE	17	-0.0982	0.2613	0.1102	0.1187
	TOBIN Q	17	0.7015	28.3920	3.5928	6.4984
	Valid N (listwise)	17				
Property, Real Estate, and Building Construction	LTDTA	28	0.1096	0.6962	0.4203	0.1572
	STDTA	28	0.0023	0.5516	0.0464	0.1031
	TDTA	28	0.1119	1.0813	0.4667	0.1921
	ROA	28	-0.0390	0.1577	0.0467	0.0490
	ROE	28	-0.0678	0.2477	0.0824	0.0791
	TOBIN Q	28	0.5771	6.5260	1.5948	1.3373
	Valid N (listwise)	28				
Infrastructure, Utilities, and Transportation	LTDTA	21	0.0030	7.0487	0.5649	1.4960
	STDTA	21	0.0405	10.2306	0.7132	2.1917
	TDTA	21	0.0436	17.2792	1.2781	3.6737
	ROA	21	-1.4122	0.0820	-0.0555	0.3170
	ROE	21	-0.1684	1.9283	0.1349	0.4272
	TOBIN Q	21	0.3848	19.2872	2.0365	4.0037
	Valid N (listwise)	21				
Trade, Services and Investment	LTDTA	54	0.0088	0.4577	0.1645	0.1160
	STDTA	54	0.0167	1.6674	0.3080	0.2782
	TDTA	54	0.0404	1.7854	0.4725	0.2926
	ROA	54	-0.5091	0.3305	0.0288	0.1124
	ROE	54	-0.7426	0.3442	0.0690	0.1755
	TOBIN Q	54	0.2927	5.9107	1.7049	1.1619
	Valid N (listwise)	54				

Table 13 Regression Analysis Family Firms by Industry Sectors (Dependent Variable ROA)

Model	Industry Sectors	Predictors	Unstandardized Coefficients		Standardized Coefficients Beta	t	Sig.	Optimum Leverage (f(x) > 0)
			B	Std. Error				
LTDTA	All Sectors in Family Firms	(Constant)	0.030	0.059		0.514	0.608	0.1519
		LTDTA	-0.199	0.100	-0.144	-1.995	0.048	
	Agriculture	(Constant)	0.101	0.148		0.684	0.543	0.2655
		STDTA	-0.381	0.426	-0.458	-0.893	0.438	
	Mining	(Constant)	-0.622	1.216		-0.511	0.617	1.6800
		STDTA	0.370	3.622	0.026	0.102	0.920	
	Basic Industry and Chemicals	(Constant)	0.062	0.014		4.481	0.000	0.4063
		STDTA	-0.152	0.054	-0.479	-2.837	0.009	
	Miscellaneous Industry	(Constant)	0.080	0.022		3.570	0.003	0.3210
		STDTA	-0.250	0.079	-0.618	-3.147	0.006	
	Consumer Goods	(Constant)	0.111	0.027		4.065	0.001	0.4512
		STDTA	-0.245	0.141	-0.408	-1.733	0.104	
	Property, Real Estate, and	(Constant)	0.099	0.025		3.927	0.001	0.7978
		STDTA	-0.124	0.056	-0.397	-2.204	0.037	
	Infrastructure, Utilities, and	(Constant)	0.060	0.020		3.084	0.006	0.2938
		STDTA	-0.205	0.012	-0.967	-16.432	0.000	
	Trade, Services and	(Constant)	0.050	0.027		1.881	0.066	0.3855
		STDTA	-0.130	0.133	-0.134	-0.979	0.332	
STDTA	All Sectors in Family Firms	(Constant)	0.031	0.057		0.554	0.580	0.1836
		STDTA	-0.171	0.069	-0.179	-2.489	0.014	
	Agriculture	(Constant)	-0.008	0.087		-0.095	0.930	0.1762
		STDTA	-0.047	0.211	-0.127	-0.222	0.839	
	Mining	(Constant)	2.318	0.843		2.749	0.015	0.1867
		STDTA	-12.420	3.198	-0.708	-3.884	0.001	
	Basic Industry and Chemicals	(Constant)	0.076	0.019		4.014	0.000	0.6100
		STDTA	-0.125	0.048	-0.449	-2.610	0.015	
	Miscellaneous Industry	(Constant)	0.090	0.038		2.396	0.029	0.6398
		STDTA	-0.140	0.078	-0.410	-1.800	0.091	
	Consumer Goods	(Constant)	0.127	0.046		2.740	0.015	0.6172
		STDTA	-0.206	0.172	-0.296	-1.201	0.248	
	Property, Real Estate, and	(Constant)	0.052	0.010		5.135	0.000	0.4674
		STDTA	-0.111	0.091	-0.233	-1.224	0.232	
	Infrastructure, Utilities, and	(Constant)	0.045	0.015		2.989	0.008	0.3211
		STDTA	-0.142	0.007	-0.979	-20.991	0.000	
	Trade, Services and	(Constant)	0.098	0.019		5.076	0.000	0.4367
		STDTA	-0.224	0.047	-0.555	-4.806	0.000	
TDTA	All Sectors in Family Firms	(Constant)	0.035	0.058		0.609	0.543	0.3539
		TDTA	-0.099	0.042	-0.170	-2.365	0.019	
	Agriculture	(Constant)	0.095	0.161		0.591	0.596	0.5121
		TDTA	-0.186	0.240	-0.408	-0.774	0.495	
	Mining	(Constant)	2.172	1.492		1.455	0.166	0.4205
		TDTA	-5.164	2.677	-0.446	-1.929	0.073	
	Basic Industry and Chemicals	(Constant)	0.111	0.018		6.153	0.000	0.7533
		TDTA	-0.147	0.031	-0.677	-4.782	0.000	
	Miscellaneous Industry	(Constant)	0.132	0.032		4.145	0.001	0.8022
		TDTA	-0.164	0.046	-0.669	-3.604	0.002	
	Consumer Goods	(Constant)	0.168	0.043		3.895	0.001	0.7031
		TDTA	-0.239	0.103	-0.515	-2.325	0.035	
	Property, Real Estate, and	(Constant)	0.100	0.022		4.457	0.000	0.8736
		TDTA	-0.115	0.045	-0.450	-2.568	0.016	
	Infrastructure, Utilities, and	(Constant)	0.052	0.016		3.308	0.004	0.6200
		TDTA	-0.084	0.004	-0.978	-20.310	0.000	
	Trade, Services and	(Constant)	0.134	0.024		5.581	0.000	0.6017
		TDTA	-0.223	0.043	-0.581	-5.142	0.000	

Further, Average value of TDTA is range between 38.2% and 127.8% for Consumer Goods Industry and Infrastructure, Utilities, and Transportation, respectively. Then, Average value of ROA is range between -51.3% and 7.6% for Mining, and Consumer Goods Industry, respectively. Moreover, average value of ROE is range between -581.7% and 22.31% for Mining and Agriculture, respectively.

Eventually, average value of Tobin's Q is range between 0.98 times and 3.59 times Agriculture and Consumer Goods Industry, respectively.

4.5.2 Industry Sectors Linear Regression

In order to find an industry benchmark for maximum leverage in 8 different sectors, regression linear analyses for each of sector was carried out to seek for a linear function of financial leverage related to ROA as shown in Table 13. A maximum leverage can be obtained from a leverage that give a positive result of dependent variable ROA ($f(x) > 0$) which can be determined from the regression linear function shown in Table 13.

Table 14 Optimum Leverage in Family Firms

Industry Sectors	LTDTA	STDTA	TDTA
All Sectors in Family Firms Group	0.1519	0.1836	0.3539
Agriculture	0.2655	0.1762	0.5121
Mining	1.6800	0.1867	0.4205
Basic Industry and Chemicals	0.4063	0.6100	0.7533
Miscellaneous Industry	0.3210	0.6398	0.8022
Consumer Goods Industry	0.4512	0.6172	0.7031
Property, Real Estate, and Building Construction	0.7978	0.4674	0.8736
Infrastrcture, Utilities, and Transportation	0.2938	0.3211	0.6200
Trade, Services and Investment	0.3855	0.4367	0.6017

4.5.3 Industry Sectors Maximum Leverage

It is suggested that the maximum leverage (LTDTA, STDTA, and TDTA) can be found from dividing an absolute number of constant coefficient and unstandardized coefficient of beta shown in Table 13.

Therefore, a maximum leverage numbers for each sector can be determined and could become a benchmark for each sector of industry for not exceeding the maximum leverage numbers to achieve for positive ROA. Table 14 shows a summary of the maximum leverage for each sector.

Maximum leverage of LTDTA is range between 26.5% and 168% for Agriculture and Mining, respectively. Subsequently, Maximum leverage of STDTA is range between 17.6% and 63.9% for Agriculture and Miscellaneous Industry, respectively. Further, Maximum leverage of TDTA is range between 42.05% and 87.36% for Mining and Property, Real Estate and Building Construction, respectively.

5. CONCLUSION

This study provides new evidence on family business research in Indonesia. The study elaborated the financial leverage effect on firm performance with contingent role on firm size. Using data from 412 listed companies listed on the IDX over the period 2016-2018, this study used three measures of financial leverage LTDTA, STDTA, and TDTA, and three measures of firm performance – ROA, ROE, and market performance Tobin's Q.

This study revealed that the non-family firms outperformed the family firms in term of accounting performance and market performance. This result support previous findings of Bambang and Hermawan (2012) and contradict with Allouche (2008), Anderson and Reeb (2003), and Maury (2006). This implies that non-family firm generate profits more effectively when compared to family firms, and non-family firms have a better exposure in market as compared to family firms.

The findings of the study also suggest that family firms tend to utilize less debt compare to non-family firms. This result support previous research study of Gallo and Vilaseca (1996), Gonzales (2013), Cleary and Alderighi (2018). This implies that family firms tend to be more risk averse and avoiding debt compare to non-family firms. Unsurprisingly, family firms tend to favor internal resources of debt to fund their business.

Subsequently, this paper found that there is a negative relationship between financial leverage and accounting performance of ROA. This result support previous evidence resulted by lbhagui and Olokoyo (2018), Gonzales (2013), Jermias and Yigit (2019), Salawu (2007), Vithessonthi and Tongurai (2015), and contradict previous study of Detthamrong (2017), Goel (2015) and Rehman (2013). This result may imply that higher leverage will lead to low rates of accounting performance ratio of ROA.

However, this study found that there is a positive relationship between financial leverage and market performance of Tobin's Q. This result support previous study of lbhagui an Olokoyo (2018) and Yang (2016). This result implies that higher leverage will lead to a better market performance.

Furthermore, this study supports our initial hypotheses which indicate the impact of financial leverage on accounting performance of ROA is contingent on firm size and changes as we move from small-size to large-size of the company.

It is suggested that the negative effect of financial leverage on accounting performance of ROA is more prominent in small-sized company. Then, it reduces the effect of the negative relationship on large-sized company. Hence, it is in line with previous research study of lbhagui and Olokoyo (2018), and Ramli (2019), and implies that the large-sized company most likely generate more profit by utilizing their debts and lead to better performance compare to small-sized company.

Further, this study result shows a positive effect of financial leverage on market performance of Tobin's Q is more prominent in small-sized company. Then, it reduces the effect of the positive relationship on large-sized company. It is supported by previous study of lbhagui and Olokoyo (2018). This result implies that small-sized company has a better market performance compare to large-sized company. However, it has to be noted that increased leverage will cause increased fragility to businesses.

Hence, this research has produced an industry benchmark of maximum leverage for 8 different industry listed in IDX. This maximum leverage can be used as a benchmark for each sector in practicing debt usage to generate a positive accounting performance of ROA. The limitation of this research is the short range of data gathered from IDX for only period of 2016-2018, longer period will give more precise and exact depiction of the company performance across economic and business cycles.

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